

# Extended sign-kill for $\widehat{\lambda} = (2, 2)$ : the complete $E_j^-$ chain at all $q \geq 2$

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## Abstract

Let  $\lambda = (2, 2, 1^q) \vdash n = q + 4$  with  $q \geq 2$ ,  $\ell = q + 2$ , and  $B = B_{\ell+1}^{(\lambda)} V_\lambda := R'_{\ell+1} \cdots R'_n V_\lambda$ . We prove the extended sign-kill conjecture for this family:

$$B \subseteq \bigcap_{j=1}^{q-1} E_j^- |_{V_\lambda} \quad \text{and} \quad B \not\subseteq E_j^- |_{V_\lambda} \quad \text{for } j \geq q.$$

Combined with the universal  $B \subseteq E_\ell^+$  (Theorem A, May-13 evening), this completely identifies the  $T_j$ -eigenspace containments of  $B$  for the family.

The proof is short. The May-13 paper `2026-05-13-non-hook-22-1n.tex` established  $\dim B = 1$  and identified the explicit 4-SYT support of the spanning vector  $b$ . We observe that this support immediately forces the column-1 entries  $1, 2, \dots, n-4$  to occupy column-1 rows  $1, 2, \dots, n-4$  in every support SYT. The Hoefsmit same-column rule gives  $T_j v_T = -v_T$  for each support SYT and  $1 \leq j \leq n-5 = q-1$ , hence  $T_j b = -b$ . Sharpness for  $j = q, q+1, \dots$  follows from the non-vanishing of the four support coefficients (also from May-13) together with the off-diagonal Hoefsmit blocks at axial distance  $\geq 2$ .

This closes the  $\widehat{\lambda} = (2, 2)$  row of the May-14 extended sign-kill conjecture table, in both directions.

## 1 Setup

We use the conventions of the May-13–May-14 series.  $H_q(S_n)$  is the Iwahori–Hecke algebra over  $\mathbb{Q}(q)$  with generators  $T_1, \dots, T_{n-1}$  and quadratic relation  $T_j^2 = (q-1)T_j + q$ .  $V_\lambda$  is the Specht module in the Hoefsmit seminormal basis  $\{v_T : T \in \text{SYT}(\lambda)\}$ .

For  $1 \leq j \leq n-1$ , set  $S_j := T_j + 1$ . The Hecke relation factors as  $S_j(T_j - q) = 0$ , equivalently  $T_j S_j = q S_j = S_j T_j$ . Write

$$E_j^+ := \ker(T_j - q) |_{V_\lambda} = \text{Im}(S_j |_{V_\lambda}), \quad E_j^- := \ker(S_j) |_{V_\lambda}.$$

Since the  $T_j$  acts with minimal polynomial  $(T - q)(T + 1)$  and the two eigenvalues are distinct (for generic  $q$ ),  $V_\lambda = E_j^+ \oplus E_j^-$ .

For  $k \geq 1$ , let  $R'_k := S_{k-1} S_{k-2} \cdots S_1$ . For  $\lambda \vdash n$  with  $\ell = \ell(\lambda)$ , set

$$\Omega = \Omega_\ell^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n \in H_q, \quad B := \Omega \cdot V_\lambda.$$

For  $\lambda = (2, 2, 1^q)$ ,  $n = q + 4$  and  $\ell = q + 2$ , so  $\Omega = R'_{q+3} R'_{q+4}$  has two  $R'$ -blocks. By the May-13 multi-corner-dimension paper,  $\dim B = f^{(1,1)} = 1$  for  $q \geq 0$  (stable regime starts at  $q = 0$  here).

**SYT labelling for  $\lambda = (2, 2, 1^q)$ .** An SYT  $T$  of  $\lambda$  has cell  $(1, 1)$  always equal to 1 and column 2 consisting of two cells  $(1, 2)$  and  $(2, 2)$ . Setting  $a := T(1, 2)$  and  $c := T(2, 2)$ , we have  $2 \leq a < c \leq n$  and  $(a, c) \neq (2, 3)$  (the latter forced by row-2 increasing). Conversely, every such pair  $(a, c)$  yields a unique SYT, denoted  $T^{(a, c)}$ , by filling column 1 at rows  $1, 2, \dots, n-2$  with the elements of  $\{1\} \cup \{2, \dots, n\} \setminus \{a, c\}$  in increasing order. Total count:  $\binom{n-1}{2} - 1 = f^\lambda$ .

## 2 Input from prior papers

**Theorem 1** (Support of  $B$ , May-13). *For  $\lambda = (2, 2, 1^q)$  with  $q \geq 2$  (i.e.  $n \geq 6$ ),*

$$B = \text{span}(b), \quad b = \sum_{(a, c) \in \mathcal{S}_n} C_{(a, c)} v_{T^{(a, c)}},$$

where

$$\mathcal{S}_n := \{(n-3, n-2), (n-3, n-1), (n-2, n), (n-1, n)\}$$

and all four coefficients  $C_{(a, c)} \in \mathbb{Q}(q)$  are nonzero.

*Proof.* This is precisely Theorem 1 (combined with Proposition 5.4 on the terminal step) of 2026-05-13-non-hook-

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**Theorem 2** (Universal  $T_\ell$  scalar, May-13 evening). *For any partition  $\lambda \vdash n$  with  $1 \leq \ell = \ell(\lambda) \leq n-1$ ,  $T_\ell \cdot \Omega^{(\lambda)} = q \cdot \Omega^{(\lambda)}$  in  $H_q$ . In particular,  $B \subseteq E_\ell^+|_{V_\lambda}$ .*

## 3 The extended sign-kill theorem

**Theorem 3** (Main). *For  $\lambda = (2, 2, 1^q)$  with  $q \geq 2$ , let  $\ell = q+2$ ,  $n = q+4$ . The image  $B = \Omega \cdot V_\lambda$  satisfies*

$$B \subseteq E_\ell^+ \cap \bigcap_{j=1}^{q-1} E_j^-|_{V_\lambda},$$

and this is sharp:

$$B \not\subseteq E_j^- \text{ for } j \in \{q, q+1, \dots, n-1\}, \quad B \not\subseteq E_j^+ \text{ for } j \in \{1, \dots, n-1\} \setminus \{\ell\}.$$

We prove the four containments and the four non-containments separately.

### 3.1 $E_\ell^+$ containment

Immediate from Theorem 2.

### 3.2 $E_j^-$ containment for $1 \leq j \leq q-1$

The key combinatorial observation:

**Lemma 4** (Column 1 structure of support SYTs). *For  $(a, c) \in \mathcal{S}_n$  with  $a < c$ , column 1 of  $T^{(a, c)}$  holds the elements of  $\{1, 2, \dots, n\} \setminus \{a, c\}$  at rows  $1, 2, \dots, n-2$  in increasing order; in particular  $T^{(a, c)}(i, 1) = i$  if and only if  $i \leq a-1$ .*

*Proof.* By definition of  $T^{(a,c)}$ . Since  $a < c$ , the sorted column-1 sequence is

$$1, 2, \dots, a-1, a+1, a+2, \dots, c-1, c+1, c+2, \dots, n.$$

The  $i$ -th element (occupying row  $i$  of column 1) equals  $i$  iff  $i \leq a-1$ . □ □

**Remark 5** (The three bounds we will use). By Lemma 4, the column-1 “identity prefix” has length  $a-1$  in  $T^{(a,c)}$ . We will repeatedly use:

- All four members of  $\mathcal{S}_n$  have  $a \geq n-3$ , giving  $T^{(a,c)}(i, 1) = i$  for  $i \leq n-4$ .
- $(n-2, n)$  and  $(n-1, n)$  have  $a \geq n-2$ , giving the additional  $T^{(a,c)}(n-3, 1) = n-3$ .
- $(n-1, n)$  has  $a = n-1$ , giving also  $T^{(a,c)}(n-2, 1) = n-2$ .

**Proposition 6** ( $E_j^-$  containment). *For  $1 \leq j \leq q-1$  and every  $(a, c) \in \mathcal{S}_n$ ,  $T_j v_{T^{(a,c)}} = -v_{T^{(a,c)}}$ . Consequently  $T_j b = -b$ , i.e.,  $b \in E_j^-$ .*

*Proof.* Let  $T := T^{(a,c)}$ . For  $1 \leq j \leq q-1 = n-5$ , both  $j$  and  $j+1$  lie in  $\{1, 2, \dots, n-4\}$ . By Lemma 4, the entries  $j$  and  $j+1$  are at cells  $(j, 1)$  and  $(j+1, 1)$  of  $T$ ; in particular they occupy the same column (column 1) at adjacent rows. By the Hoefsmit seminormal action (same-column case),  $T_j v_T = -v_T$ .

By linearity over  $(a, c) \in \mathcal{S}_n$  in  $b = \sum_{(a,c)} C_{(a,c)} v_{T^{(a,c)}}$  (Theorem 1),  $T_j b = -b$ . □ □

### 3.3 $E_j^-$ non-containment for $q \leq j \leq n-1$

The argument uses non-vanishing of the four support coefficients together with the existence of an off-diagonal Hoefsmit block.

**Lemma 7** (Off-diagonal action at  $j = q$ ). *For  $j = q$  (so  $j+1 = q+1$ ) and the support SYT  $T^* := T^{(n-3, n-2)} = T^{(q+1, q+2)}$ , the entries  $q$  and  $q+1$  sit at cells  $(q, 1)$  and  $(1, 2)$  of  $T^*$  respectively (different row and different column). The corresponding axial distance is  $d = c(1, 2) - c(q, 1) = 1 - (1 - q) = q$ . In particular, the Hoefsmit  $T_q$ -action on the 2-block  $\{v_{T^*}, v_{s_q T^*}\}$  is non-degenerate over  $\mathbb{Q}(q)$ , i.e.,  $T_q v_{T^*} = \alpha v_{T^*} + \beta v_{s_q T^*}$  with  $\beta \neq 0$ .*

*Here  $s_q T^* = T^{(q, q+2)}$  (swap entries  $q$  and  $q+1$ , which exchanges  $T^*(q, 1) = q$  with  $T^*(1, 2) = q+1$ ); this is a valid SYT of  $\lambda$  and is not an element of  $\mathcal{S}_n$ .*

*Proof.* Lemma 4 gives  $T^*(q, 1) = q$ . By definition of  $T^* = T^{(q+1, q+2)}$ ,  $T^*(1, 2) = q+1$ . So  $j = q$  at  $(q, 1)$  (content  $1-q$ ) and  $j+1 = q+1$  at  $(1, 2)$  (content 1). Different row and column, axial distance  $d = 1 - (1 - q) = q$ . For  $q \geq 2$  ( $\Leftrightarrow |d| \geq 2$ ), the Hoefsmit ratio  $\tau(d) = (q^{|d|} - 1)/(q^d \cdot (q-1)) \cdot (\cdot)$  (the precise formula does not matter) gives  $\beta \neq 0$  over  $\mathbb{Q}(q)$ . Concretely, the off-diagonal entry of the standard Hoefsmit 2-block at axial distance  $d$  is a nonzero rational function in  $q$  for all  $|d| \geq 2$ .

Swapping  $q$  and  $q+1$  in  $T^*$  yields the SYT  $T^{(q, q+2)}$  (with column-2 entries  $q, q+2$ ). Since  $q < q+1$  and  $(q, q+2) \notin \mathcal{S}_n$  (which only contains pairs with both entries  $\geq n-3 = q+1$ ), this swapped SYT lies outside the support of  $b$ . □ □

**Proposition 8** ( $B \not\subseteq E_q^-$ ).  $T_q b \neq -b$ , hence  $b \notin E_q^-$ .

*Proof.* By Theorem 1,  $b = \sum_{(a,c) \in \mathcal{S}_n} C_{(a,c)} v_{T^{(a,c)}}$  with all four  $C_{(a,c)} \neq 0$ . Consider the coefficient of  $v_{T^{(q, q+2)}}$  in  $T_q b$ . This SYT is *not* in  $\mathcal{S}_n$ , so it is not in the support of  $b$ ; hence the coefficient of  $v_{T^{(q, q+2)}}$  in  $-b$  is 0. On the other hand, by Lemma 7,  $T_q v_{T^*}$  produces the term  $\beta v_{T^{(q, q+2)}}$  with  $\beta \neq 0$ . We need to check that no other summand of  $T_q b$  cancels this contribution.

A summand  $T_q(C_{(a,c)} v_{T^{(a,c)}})$  produces a term proportional to  $v_{T^{(q, q+2)}}$  only if either

- $T^{(a,c)} = T^{(q,q+2)}$  (impossible, since  $T^{(q,q+2)} \notin \mathcal{S}_n$ ),
- or  $s_q T^{(a,c)} = T^{(q,q+2)}$ .

$s_q T^{(a,c)}$  swaps the cells of  $q$  and  $q+1$  in  $T^{(a,c)}$ . The resulting SYT has the same column-2 entries as  $T^{(a,c)}$  except that any occurrence of  $q$  or  $q+1$  in column 2 is swapped.

Among the four  $(a, c) \in \mathcal{S}_n$ :

- $(a, c) = (q+1, q+2)$ : column 2 entries are  $q+1, q+2$ , both in column 2. Swap of  $q, q+1$  moves  $q+1$  from  $(1, 2)$  to  $(q, 1)$  and  $q$  from  $(q, 1)$  to  $(1, 2)$  (provided  $T^{(a,c)}(q, 1) = q$ , which Lemma 4 gives). Resulting SYT: column-2 entries  $q, q+2$ , i.e.,  $T^{(q,q+2)}$ . *This is the only contributor.*
- $(a, c) = (q+1, q+3)$ : same as above for the position of  $q+1$  at  $(1, 2)$ . Swap  $q \leftrightarrow q+1$ : now  $q$  at  $(1, 2)$ ,  $q+1$  at  $(q, 1)$ . Column-2 entries:  $q, q+3$ , i.e.,  $T^{(q,q+3)}$ . *Different from  $T^{(q,q+2)}$ .*
- $(a, c) = (q+2, q+4)$ : here  $a = q+2 \geq n-2$ , so by Lemma 4 the identity prefix has length  $a-1 = q+1$ , giving  $T^{(a,c)}(q, 1) = q$  and  $T^{(a,c)}(q+1, 1) = q+1$ . Hence  $q$  and  $q+1$  are in column 1 at adjacent rows, so  $T_q v_{T^{(a,c)}} = -v_{T^{(a,c)}}$  (same-column case). *No off-diagonal contribution.*
- $(a, c) = (q+3, q+4)$ : here  $a = q+3 = n-1$ , identity prefix length  $q+2$ , so analogously  $T_q v_{T^{(a,c)}} = -v_{T^{(a,c)}}$ .

Therefore the coefficient of  $v_{T^{(q,q+2)}}$  in  $T_q b$  equals  $C_{(q+1,q+2)} \beta \neq 0$ , while the same coefficient in  $-b$  is 0. Hence  $T_q b \neq -b$ , so  $b \notin E_q^-$ .  $\square$

**Proposition 9** ( $B \not\subseteq E_j^-$  for  $q+1 \leq j \leq n-1$ ). *For each  $j \in \{q+1, q+2, \dots, n-1\}$ ,  $T_j b \neq -b$ .*

*Proof.* We treat the three sub-ranges separately.

$j = q+1 = \ell-1 = n-3$ . Positions of  $j, j+1$  in each support SYT  $T^{(a,c)}$ :

- $(a, c) = (n-3, n-2)$ :  $j = a$  at  $(1, 2)$ ,  $j+1 = c$  at  $(2, 2)$ . *Same column:*  $T_j v = -v$ .
- $(a, c) = (n-3, n-1)$ :  $j = a$  at  $(1, 2)$ ,  $j+1 = n-2$  in column 1 (specifically at row  $n-3$ ). *Off-diagonal*, axial distance  $1 - (-(n-4)) = n-3$ .
- $(a, c) = (n-2, n)$ :  $j = n-3$  in column 1 at row  $n-3$ ,  $j+1 = a = n-2$  at  $(1, 2)$ . *Off-diagonal*, axial distance  $-(n-3)$ .
- $(a, c) = (n-1, n)$ :  $j, j+1 = n-3, n-2$  both in column 1 at rows  $n-3, n-2$ . *Same column:*  $T_j v = -v$ .

Swap of  $j, j+1$  in  $T^{(n-3,n-1)}$ :  $j = n-3$  moves from  $(1, 2)$  to column 1 row  $n-3$ ;  $j+1 = n-2$  moves from column 1 row  $n-3$  to  $(1, 2)$ . Resulting SYT: column 2 entries  $(n-2, n-1)$ , i.e.  $T^{(n-2,n-1)}$ , which is outside  $\mathcal{S}_n$ . Swap of  $j, j+1$  in  $T^{(n-2,n)}$ : analogously yields  $T^{(n-3,n)}$ , also outside  $\mathcal{S}_n$ . Since these two swap-targets are distinct from each other and both have  $|d| = n-3 \geq 3$  (non-degenerate Hoefsmit block), each contributes a nonzero coefficient to  $T_{q+1} b$  on a SYT outside  $\mathcal{S}_n$ , while  $-b$  is supported entirely in  $\mathcal{S}_n$ . Hence  $T_{q+1} b \neq -b$ .

$j = \ell = n-2$ . By Theorem 2,  $T_\ell b = qb$ . For generic  $q \neq -1$ ,  $qb \neq -b$ , so  $b \notin E_\ell^-$ .

$j = \ell + 1 = n - 1$ . Positions of  $j, j + 1 = n - 1, n$  in each support SYT:

- $(n - 3, n - 2)$ : column 1 entries are  $\{1, 2, \dots, n - 4, n - 1, n\}$  placed at rows  $1, \dots, n - 2$  in increasing order, so  $n - 1$  at row  $n - 3$  and  $n$  at row  $n - 2$  of column 1. *Same column*:  $T_{n-1}v = -v$ .
- $(n - 3, n - 1)$ : column 1 omits  $n - 3$  and  $n - 1$ , so column 1 =  $\{1, \dots, n - 4, n - 2, n\}$  at rows  $1, \dots, n - 2$ .  $n - 1$  at  $(2, 2)$ ,  $n$  at row  $n - 2$  of column 1. Different row, column. Off-diagonal.
- $(n - 2, n)$ : column 1 =  $\{1, \dots, n - 3, n - 1\}$ .  $n - 1$  at row  $n - 2$  of column 1,  $n = c$  at  $(2, 2)$ . Off-diagonal.
- $(n - 1, n)$ :  $n - 1, n$  at  $(1, 2), (2, 2)$ . Same column 2.  $T_{n-1}v = -v$ .

Swap of  $j, j + 1$  in  $T^{(n-3, n-1)}$ :  $n - 1$  moves from  $(2, 2)$  to column 1 row  $n - 2$ ;  $n$  moves from column 1 row  $n - 2$  to  $(2, 2)$ . Resulting SYT: column 2 entries  $(n - 3, n)$ , i.e.  $T^{(n-3, n)}$ , outside  $\mathcal{S}_n$ . Swap in  $T^{(n-2, n)}$ : analogously yields  $T^{(n-2, n-1)}$ , also outside  $\mathcal{S}_n$ . Same argument as in the  $j = q + 1$  case: each off-diagonal contribution gives a nonzero coefficient on a distinct SYT outside the support of  $b$ . Hence  $T_{n-1}b \neq -b$ .  $\square$   $\square$

### 3.4 $E_j^+$ non-containment for $j \neq \ell$

**Proposition 10** ( $B \not\subseteq E_j^+$  for  $j \neq \ell$ ). For  $j \in \{1, \dots, n - 1\} \setminus \{\ell\}$ ,  $T_j b \neq qb$ .

*Proof.* For  $j \leq q - 1$ ,  $T_j b = -b$  by Proposition 6, which equals  $qb$  iff  $-1 = q$ , impossible for generic  $q$ . For  $j \geq q$ , the same off-diagonal arguments of Propositions 8–9 show that  $T_j b$  has a nonzero component on an SYT outside  $\mathcal{S}_n$ ; but  $qb$  is supported on  $\mathcal{S}_n$ . Hence  $T_j b \neq qb$ .  $\square$   $\square$

This completes the proof of Theorem 3.

## 4 Computational verification

**Theorem 11** (Verification). *Theorem 3, together with the support description of Theorem 1, was verified computationally at  $q = 2, 3, 4, 5, 6, 7$  (i.e.,  $n = 6, \dots, 11$ ) mod  $P = 100003$  at  $q = 11$ . Across these six cases:*

- $\dim B = 1$  in all cases.
- Support of  $b$  matches  $\mathcal{S}_n$  exactly, all four coefficients nonzero.
- $T_j b = -b$  for  $1 \leq j \leq q - 1$  in all cases.
- $T_q b \neq -b$  in all cases (verified by recovering a nonzero coefficient on  $T^{(q, q+2)}$ ).
- $T_\ell b = qb$  in all cases (Theorem 2).

*Proof.* Scripts `~/projects/scratch/2026-05-14-extended-sign-kill-proof/verify_support_argument.py` and `check_S10mega_zero.py`. The scripts construct the Hoefsmit  $T_i$  matrices for  $\lambda = (2, 2, 1^q)$ , build  $\Omega$  explicitly, and check all of the above by direct linear algebra.  $\square$   $\square$

## 5 Discussion

### 5.1 What is new

The extended sign-kill conjecture (May-14 evening) asserts, for  $\lambda = (\widehat{\lambda}, 1^q)$  with  $\ell(\widehat{\lambda}) \geq 2$ ,

$$B \subseteq E_j^-|_{V_\lambda} \iff j \leq \tau(\lambda),$$

where  $\tau(\lambda) = q + 3 - \hat{\lambda}_1 - \hat{\lambda}_2$ . For  $\hat{\lambda} = (2, 2)$ ,  $\tau = q - 1$ .

The May-13 paper `2026-05-13-non-hook-22-1n.tex` proves the  $j = 1$  case (containment in  $E_1^-$ ). The May-20 paper proves the analogous  $j = 1$  case for  $\hat{\lambda} = (3, 2)$  at  $q \geq 3$ . Neither paper addresses  $j \geq 2$ .

The present paper closes the  $\hat{\lambda} = (2, 2)$  row of the conjecture in both directions:  $B \subseteq E_j^-$  for  $j \leq q - 1$ , and  $B \not\subseteq E_j^-$  for  $j \geq q$ . The proof is short and extracted directly from the support structure already in May-13; the contribution is the observation that the support immediately forces  $T(i, 1) = i$  for  $i \leq n - 4$ , plus the sharpness argument using the non-vanishing of the four support coefficients.

## 5.2 The general support rule

The argument generalizes mechanically. Define, for any  $H_q(S_n)$ -submodule  $W \subseteq V_\lambda$ ,

$$\text{supp}(W) := \{T \in \text{SYT}(\lambda) : \text{some } w \in W \text{ has nonzero coefficient on } v_T\},$$

i.e., the union of the seminormal supports of all vectors in  $W$ .

**Proposition 12** (Support rule). *Suppose  $W \subseteq V_\lambda$ . If every  $T \in \text{supp}(W)$  has  $j$  and  $j + 1$  in the same column of  $T$ , then  $W \subseteq E_j^-|_{V_\lambda}$ .*

*Proof.* Let  $w \in W$ ,  $w = \sum_{T \in \text{supp}(W)} c_T v_T$  (with some  $c_T$  possibly zero). By the Hoefsmit same-column rule,  $T_j v_T = -v_T$  for each  $T$  in the support of  $w$ . Linearity gives  $T_j w = -w$ .  $\square$   $\square$

This is the crux of the proof for  $\hat{\lambda} = (2, 2)$  above (where the support has  $T(i, 1) = i$  for  $i \leq n - 4$ , so  $j, j + 1$  are column-1-adjacent for  $j \leq n - 5 = q - 1$ ).

## 5.3 Empirical extension to $\hat{\lambda} = (3, 2)$

Computational verification (mod  $P = 100\,003$  at  $q = 11$ ) for  $\lambda = (3, 2, 1^q)$  at  $q = 3, 4, 5, 6$ :

- $\dim B = 2$  in all four cases, and  $|\text{supp}(B)| = 26$  in all four cases.
- Every  $T \in \text{supp}(B)$  has column-1 identity prefix of length  $\geq q - 1$ , i.e.,  $T(i, 1) = i$  for  $i = 1, \dots, q - 1$ . Witnesses: SYTs with  $T(q, 1) \neq q$  (specifically  $T(q, 1) = q + 1$ ) achieve the minimum prefix  $q - 1$ .
- Hence by Proposition 12,  $B \subseteq E_j^-$  for  $j \leq q - 2$  — exactly the predicted threshold  $\tau = q + 3 - 3 - 2 = q - 2$ .
- Sharpness ( $B \not\subseteq E_j^-$  for  $j \geq q - 1$ ) is also computationally verified: the off-diagonal swap-targets at  $j = q - 1$  produce SYTs outside the support, by the same argument as Section 3.

The structural proof would require characterizing  $\text{supp}(B)$  explicitly for  $(3, 2, 1^q)$ . The May-20 paper constructs  $B$  as the image of the three-branch decomposition under  $R'_{\ell+1} \cdots R'_{n-1}$ , and the resulting support is the 26-element set above. Tracing the construction to extract the support description would close the  $\hat{\lambda} = (3, 2)$  row of the conjecture. We leave this as future work.

## 5.4 Extension blueprint for general $\hat{\lambda}$

The pattern that emerges:

**Conjecture 13** (Support structure conjecture). *For  $\lambda = (\widehat{\lambda}, 1^q) \vdash n$  with  $\ell(\widehat{\lambda}) \geq 2$ , the seminormal support  $\text{supp}(B_{\ell+1}^{(\lambda)} V_\lambda)$  consists exactly of those SYTs  $T$  of  $\lambda$  satisfying  $T(i, 1) = i$  for  $i \leq q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$ .*

Conjecture 13, combined with Proposition 12, would prove the full extended sign-kill conjecture (the “ $\subseteq$ ” direction). Sharpness in each case would follow from witness SYTs at the boundary prefix length, analogously to Section 3.

The top-two-rows independence  $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$  would then have a clean interpretation: it is the minimum identity-prefix length in the support of  $B$ , and this minimum is determined by the first two rows of the column-1-deletion shape  $\widehat{\lambda}^{(\geq 2)}$  (with  $|\widehat{\lambda}_{\leq 2}^{(\geq 2)}| = \widehat{\lambda}_1 + \widehat{\lambda}_2 - 2$ ).

## 5.5 Containment versus identity

Theorem 3 gives  $B \subseteq E_\ell^+ \cap \bigcap_{j=1}^{q-1} E_j^-$ . The intersection on the right is computationally much larger than  $\dim B = 1$  (e.g., dimension 2 for  $\lambda = (2, 2, 1^q)$  at all tested  $q \geq 2$ , see May-14 evening paper’s Table 2). So the eigenspace constraints do not pin down  $B$  exactly; the closed-form  $\dim B = f^{\lambda^{(\geq 2)}} = 1$  is a sharper fact, capturing structure not detected by  $T_j$ -eigenspaces alone.

## 5.6 Status of the broader iso programme

This result closes one row of the May-14 evening conjecture table but does not affect the broader iso programme (the May-13–May-14 attempts to identify  $B$  as a known representation-theoretic object). Those attempts were closed off by the May-14 morning and afternoon refutations. The present contribution is, in effect, a “cleaning up” of the extended sign-kill empirical observation into a structurally established theorem at the simplest non-hook shape.

## 6 Status

**Established.**

- Theorem 3: complete characterization of  $T_j$ -eigenspace containments of  $B$  for  $\lambda = (2, 2, 1^q)$ ,  $q \geq 2$ .
- Sharpness for both  $E_j^-$  and  $E_j^+$  containments.

**Open.**

- Extended sign-kill at  $\widehat{\lambda} = (3, 2)$  at  $j \geq 2$  (requires May-20-style support description; computationally consistent with threshold  $\tau = q - 2$ ).
- Extended sign-kill at general  $\widehat{\lambda}$  (requires support description for general multi-corner shapes).
- Structural mechanism for the top-two-rows independence  $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$  (independent of lower rows of  $\widehat{\lambda}$ ).

**Push status.** TBD — depends on PAT refresh. Prior unpushed-commit count on `clio-vega/proofs`: 64 (May-14 evening session).

**Scripts.** All computations: `~/projects/scratch/2026-05-14-extended-sign-kill-proof/`.

- `explore_22_family.py` — ranks and intersections of  $E_j^\pm \cap B$  for  $\lambda = (2, 2, 1^q)$  at  $q = 2, 3, 4$ .
- `identify_22.py` — explicit basis vector of  $B$  in Hoefsmit basis, support count, normalized coefficients.
- `check_S10mega_zero.py` — verifies  $S_1\Omega = 0$  as a matrix on  $V_\lambda$  for  $q \geq 2$  in the  $(2, 2, 1^*)$  family, and diagnostic checks on other families  $((3, 2, 1^*), \text{etc.})$ .
- `verify_support_argument.py` — end-to-end verification of Theorem 3: support matches  $\mathcal{S}_n$ , the column-1 pattern, and the  $T_j$ -action at  $j \leq q - 1$  and  $j = q$ .